A defensible quantum allocation decision

Keep or tilt? Sector choices for the German public Kapitalrente satellite sleeve

A verified classical answer key, two QAOA representations, stress testing, holdout evaluation, and a human-readable Keep / Switch / Review decision.

Important: quantum results use IBM's local Qiskit Aer simulator. No quantum advantage is claimed.

1. The decision

At 31 December 2019, an investment committee must decide which three of six industries should form an equal-weight satellite tilt. The current holding is NoDur + Hlth + Utils; the proposal replaces NoDur with HiTec.

Profile	Decision	Portfolio	Reason
Growth	SWITCH	HiTec + Hlth + Utils	Stable across supplied stresses
Balanced	REVIEW	Base case: HiTec + Hlth + Utils	Flips under HiTec return stress
Risk-sensitive	KEEP	NoDur + Hlth + Utils	Risk rule and current tilt dominate

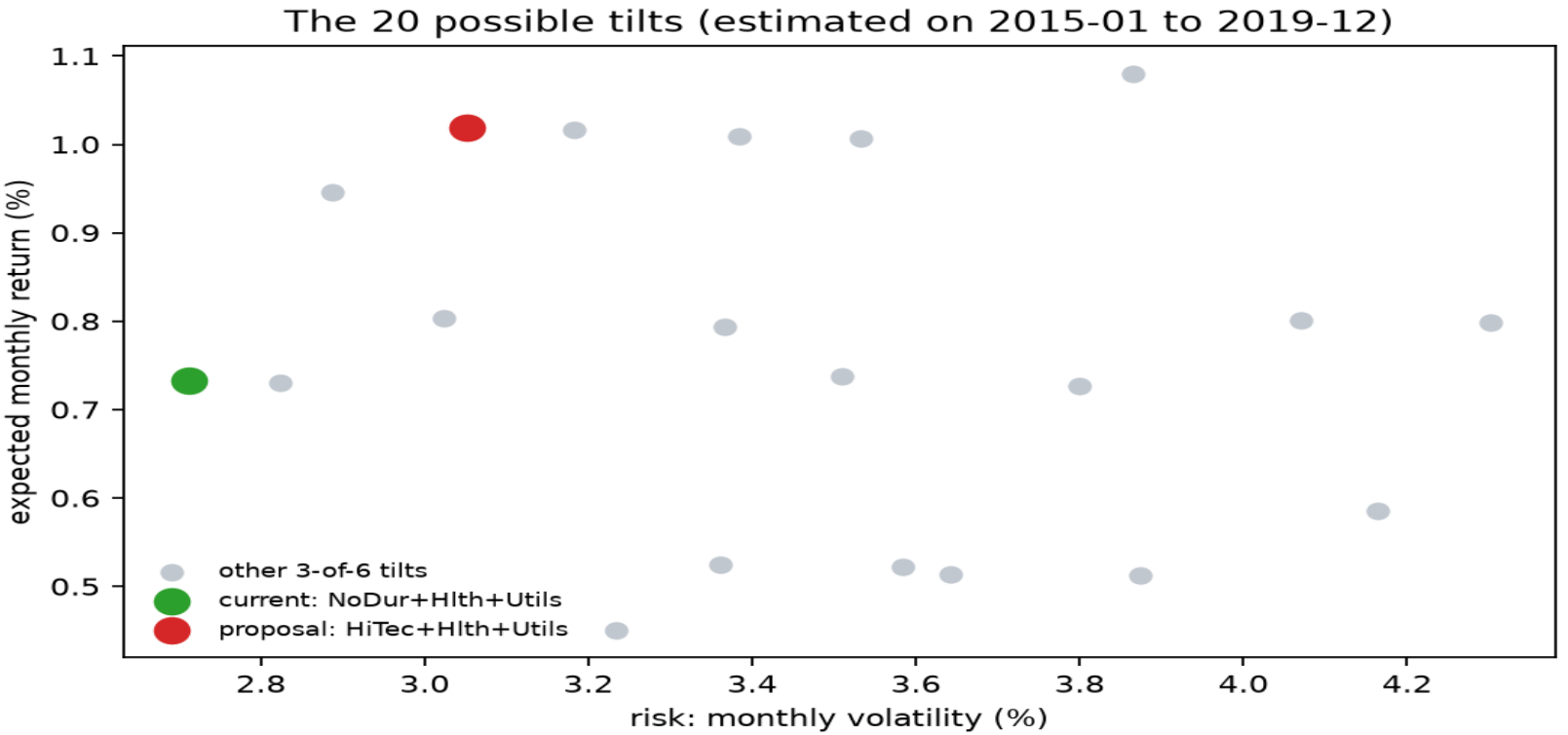
2. Data and model boundaries

Item	Treatment
Estimation window	60 monthly observations, Jan 2015 – Dec 2019
Holdout window	Jan 2020 – Dec 2024; never used for the decision
Universe	NoDur, Manuf, Enrgy, HiTec, Hlth, Utils
Decision space	20 equal-weight portfolios: choose exactly 3 of 6
Binding rules	Exactly 3 holdings and mandate loss tolerance
Preferences	Return, variance, switching penalty, soft HiTec cap

The model is a challenge-sized satellite sleeve, not the complete pension portfolio. Mandate parameters are teaching assumptions, not legal requirements.

3. Exact classical answer key

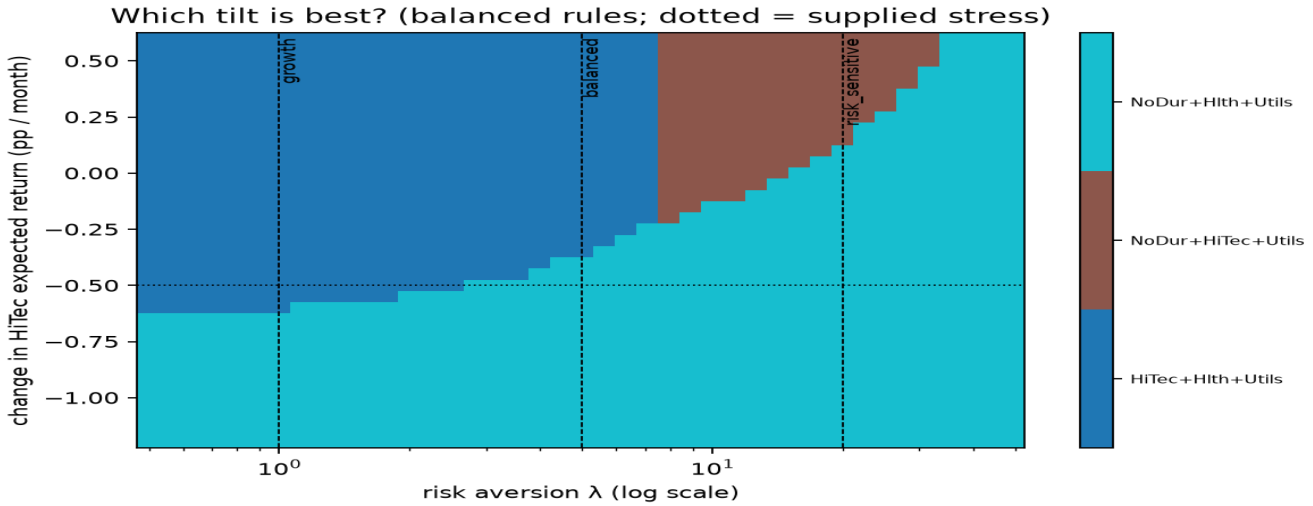
There are only C(6,3) = 20 candidate portfolios. We enumerate all of them, apply the objective and loss rule exactly, and use the result as the audit key for every quantum run.



4. Why balanced is a Review case

Scenario	Growth	Balanced	Risk-sensitive
Base	HiTec + Hlth + Utils	HiTec + Hlth + Utils	NoDur + Hlth + Utils
HiTec return stress	HiTec + Hlth + Utils	NoDur + Hlth + Utils	NoDur + Hlth + Utils
Covariance stress	HiTec + Hlth + Utils	HiTec + Hlth + Utils	NoDur + Hlth + Utils

The balanced recommendation changes under a plausible technology return stress. That is not a failed model: it is evidence that the committee should review its belief about technology's future edge before switching.



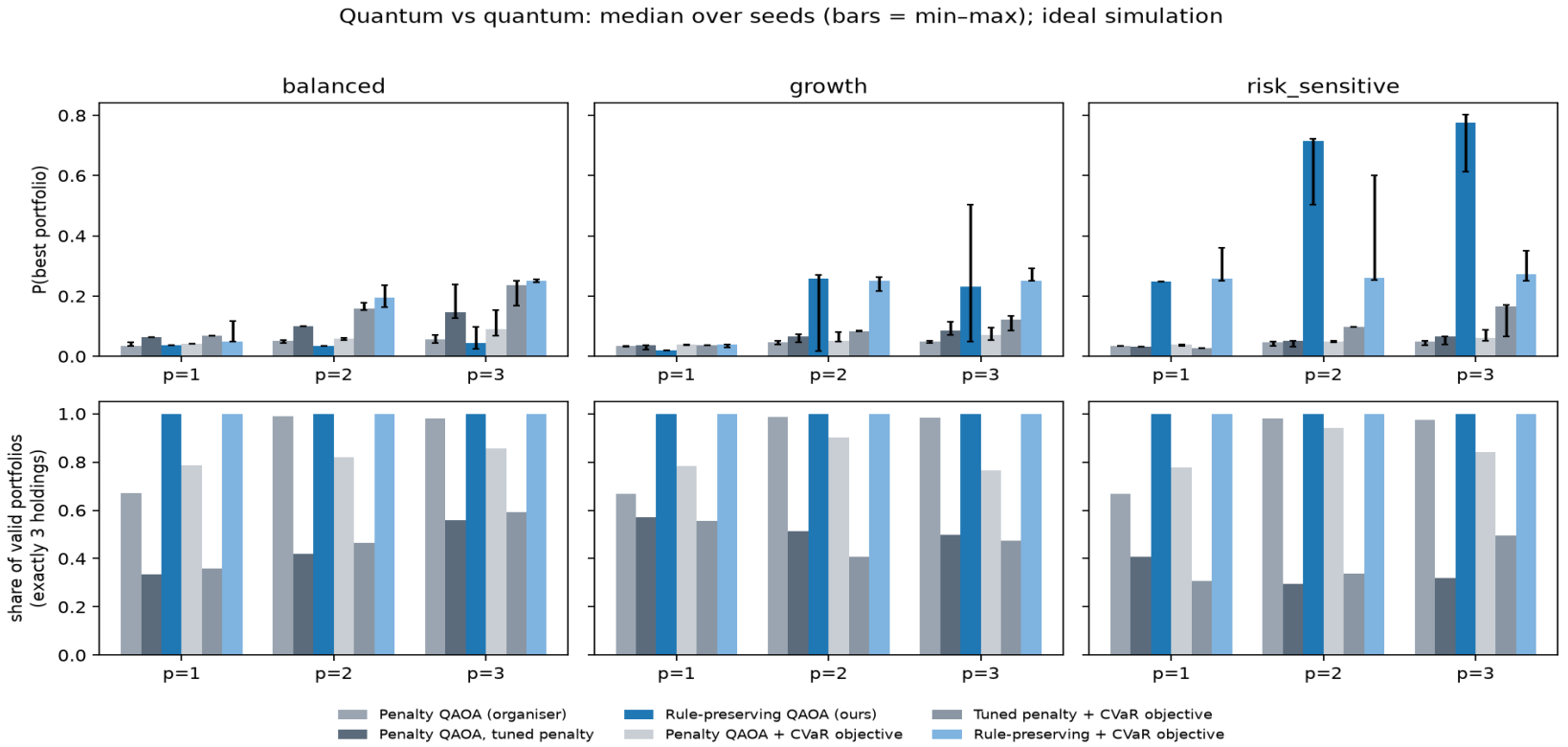
5. Where quantum fits

We encode one binary variable per industry. A bitstring identifies the selected industries. The QAOA cost combines return, risk, switching, and the soft concentration preference.

Baseline	Rule-aware design
X mixer with cardinality penalties	Dicke-state initialization
Invalid bitstrings can be sampled	XY mixer preserves exactly three holdings
Validity checked after measurement	Cardinality preserved structurally
Compared against exact enumeration	Compared against exact enumeration

The circuits are simulated with IBM's local Qiskit Aer statevector simulator and shot-based sampling. This is not IBM Quantum cloud hardware execution.

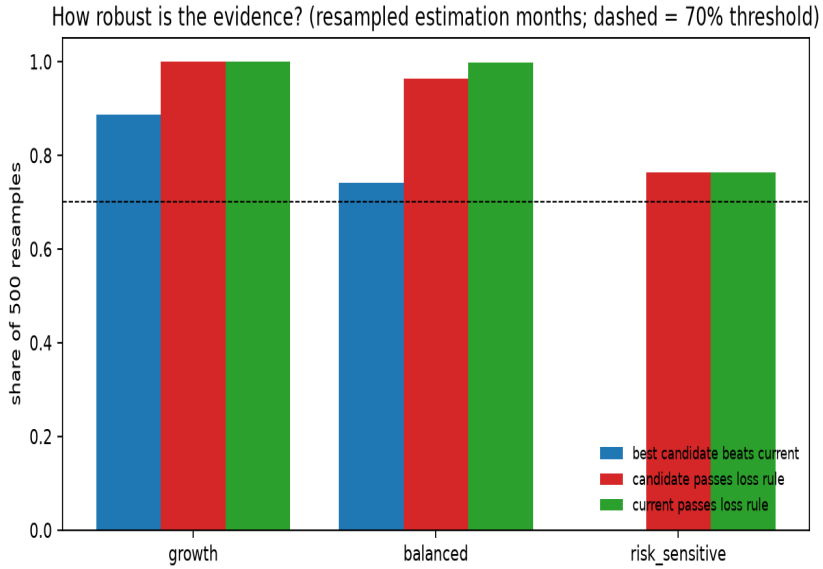
6. Quantum results

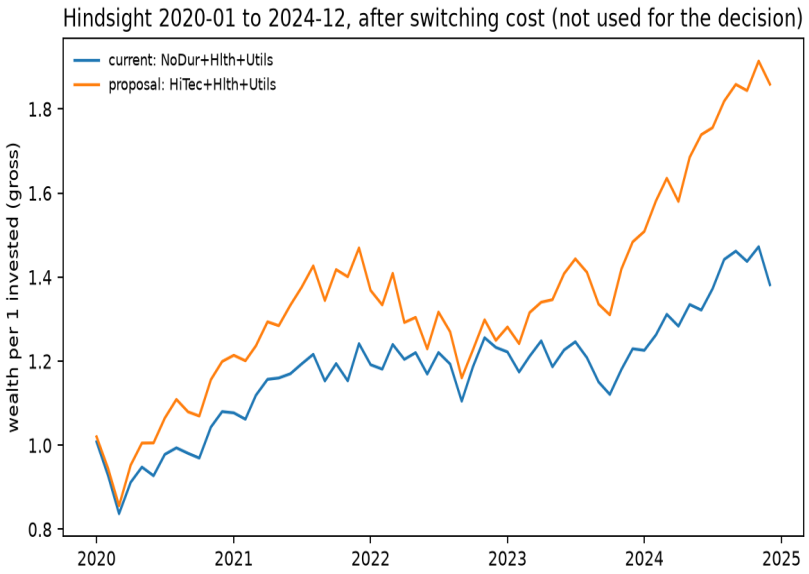


Exact enumeration remains the verifier. We report optimum probability, valid-sample fraction, seed variation, and shot noise rather than claiming a speedup.

7. Robustness and hindsight

The decision is tested under supplied return and covariance stresses, parameter changes, bootstrap resampling, and the held-out 2020–2024 period. The holdout is reported after the decision and is never used to select a portfolio.




Hindsight 2020-01 to 2024-12, after switching cost (not used for the decision)
current: NoDur+Hlth+Utils
proposal: HiTec+Hlth+Utils
wealth per 1 invested (gross)
0.8
1.0
1.2
1.4
1.6
1.8
2020
2021
2022
2023
2024
2025

8. Responsible conclusion

Claim we make	Claim we do not make
The rule-aware QAOA encoding is checked against an exact answer key.	Quantum advantage over classical enumeration
The balanced decision is stress-sensitive and should be reviewed.	Historical returns are forecasts
The holdout period exposes model and estimation risk.	This is a complete pension-portfolio recommendation
A human investment committee remains responsible.	The simulator is IBM hardware

Final message: Growth → Switch. Balanced → Review. Risk-sensitive → Keep. The value is transparent decision support, not a claim that quantum computing is already faster.

9. Demo flow

- Open the Streamlit dashboard and select a mandate.
- Show the exact classical answer key and the Keep / Switch / Review framing.
- Point to the risk/return map and current-versus-recommended comparison.
- Show the scenario flip for Balanced.
- Enable IBM Aer QAOA and compare X-penalty with Dicke + XY.
- End with the holdout chart and the explicit limitation: no quantum advantage claim.

Dashboard: http://127.0.0.1:8501

Repository: quantum-hackathon-finance / main